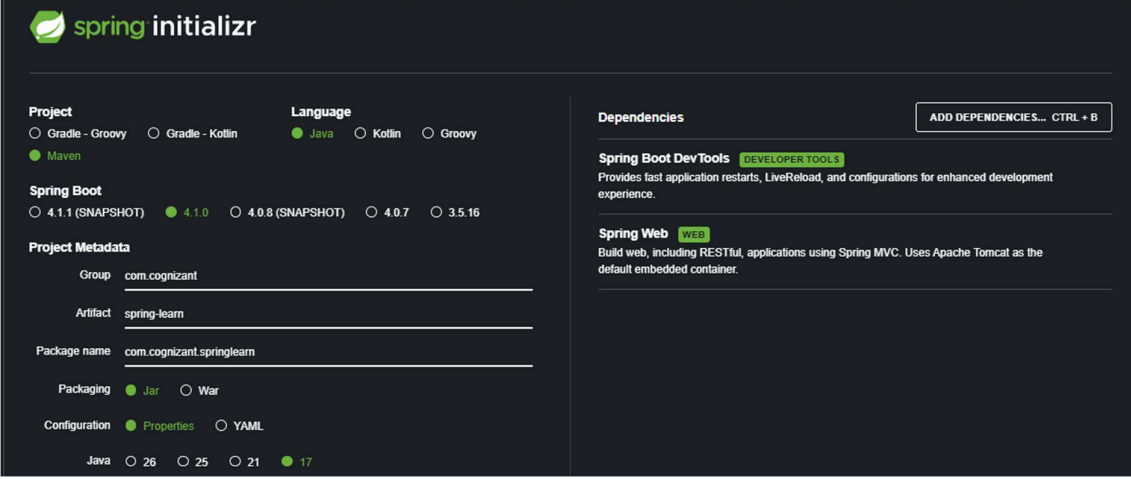


Ex 1: Create a Spring Web Project using Maven

Spring initializer:



The Spring Initializr web interface is shown with the following configuration:

- Project:** ☒ Maven
- Language:** ☒ Java
- Spring Boot:** ☒ 4.1.0
- Project Metadata:**
 - Group:
 - Artifact:
 - Package name:
 - Packaging: ☒ Jar
 - Configuration: ☒ Properties
 - Java: ☐ 26 ☐ 25 ☐ 21 ☒ 17
- Dependencies:**
 - Spring Boot DevTools** (DEVELOPER TOOLS): Provides fast application restarts, LiveReload, and configurations for enhanced development experience.
 - Spring Web** (WEB): Build web, including RESTful, applications using Spring MVC. Uses Apache Tomcat as the default embedded container.

Output after building:

```
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 3.700 s -- in com.cognizant.springlearn.SpringLearnApplicationTests
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ spring-learn ---
[INFO] Building jar: C:\Users\anjan\workspace\spring-learn\target\spring-learn-0.0.1-SNAPSHOT.jar
[INFO]
[INFO] --- spring-boot:4.1.0:repackage (repackage) @ spring-learn ---
[INFO] Replacing main artifact C:\Users\anjan\workspace\spring-learn\target\spring-learn-0.0.1-SNAPSHOT.jar with repackaged archive, adding nested dependencies in BOOT-INF/.
[INFO] The original artifact has been renamed to C:\Users\anjan\workspace\spring-learn\target\spring-learn-0.0.1-SNAPSHOT.jar.original
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 02:00 min
[INFO] Finished at: 2026-07-13T16:04:57+05:30
[INFO]
```

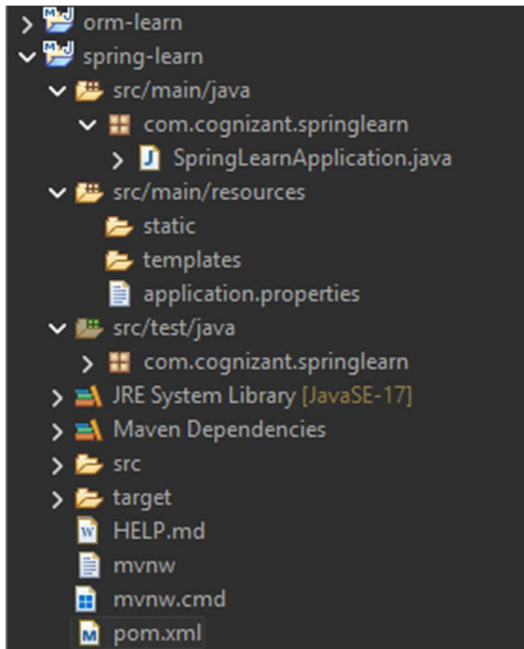
SpringLearnApplication.java:

```
1 package com.cognizant.springlearn;  
2  
3 import org.slf4j.Logger;  
4 import org.slf4j.LoggerFactory;  
5 import org.springframework.boot.SpringApplication;  
6 import org.springframework.boot.autoconfigure.SpringBootApplication;  
7  
8 @SpringBootApplication  
9 public class SpringLearnApplication {  
10  
11     private static final Logger LOGGER = LoggerFactory.getLogger(SpringLearnApplication.class);  
12  
13     public static void main(String[] args) {  
14  
15         LOGGER.info("Inside main() method - Application is starting...");  
16  
17         SpringApplication.run(SpringLearnApplication.class, args);  
18  
19         LOGGER.info("Spring Boot application started successfully.");  
20  
21     }  
22 }
```

Output:

```
16:17:37.491 [main] INFO com.cognizant.springlearn.SpringLearnApplication -- Inside main() method - Application is starting...  
16:17:37.775 [restartedMain] INFO com.cognizant.springlearn.SpringLearnApplication -- Inside main() method - Application is starting...  
  
  ____ _  
 / ___ \| | | |  
/ /___ \| |_| |  
 \___ \|____|_|_|_|  
  _____  
  :: Spring Boot ::  
  (v4.1.0)  
  
2026-07-13T16:17:38.401+05:30 INFO 27488 --- [spring-learn] [ restartedMain] c.c.springlearn.SpringLearnApplication : Starting SpringLearnApplication using Java 17.0.19 with PID 27488 (C:\  
2026-07-13T16:17:38.410+05:30 INFO 27488 --- [spring-learn] [ restartedMain] c.c.springlearn.SpringLearnApplication : No active profile set, falling back to 1 default profile: "default"  
2026-07-13T16:17:38.540+05:30 INFO 27488 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.add-properties' to 'true' to enable  
2026-07-13T16:17:38.541+05:30 INFO 27488 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level.org.springframework.boot.autoconfigure.web' property  
2026-07-13T16:17:39.851+05:30 INFO 27488 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)  
2026-07-13T16:17:39.869+05:30 INFO 27488 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]  
2026-07-13T16:17:39.869+05:30 INFO 27488 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]  
2026-07-13T16:17:39.937+05:30 INFO 27488 --- [spring-learn] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 1390 ms  
2026-07-13T16:17:40.577+05:30 INFO 27488 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'  
2026-07-13T16:17:40.590+05:30 INFO 27488 --- [spring-learn] [ restartedMain] c.c.springlearn.SpringLearnApplication : Started SpringLearnApplication in 2.806 seconds (process running for 3.206 seconds)  
2026-07-13T16:17:40.598+05:30 INFO 27488 --- [spring-learn] [ restartedMain] c.c.springlearn.SpringLearnApplication : Spring Boot application started successfully.
```

Project Folder:



pom.xml dependency hierarchy:

```
32 <dependencies>
33   <dependency>
34     <groupId>org.springframework.boot</groupId>
35     <artifactId>spring-boot-starter-webmvc</artifactId>
36   </dependency>
37
38   <dependency>
39     <groupId>org.springframework.boot</groupId>
40     <artifactId>spring-boot-devtools</artifactId>
41     <scope>runtime</scope>
42     <optional>true</optional>
43   </dependency>
44   <dependency>
45     <groupId>org.springframework.boot</groupId>
46     <artifactId>spring-boot-starter-webmvc-test</artifactId>
47     <scope>test</scope>
48   </dependency>
49 </dependencies>
```